

# .XIA

## **Introduction**

To what extent can playing an educational game help people struggling with Dyslexia? In order to answer this question By using the Unity game engine and employing C# in order to lay the foundation for the game, we can also employ the use of open source game assets such as sprites so as to create a lightweight, cross-platform and intuitive game which could be accessible to kids without being limited by device.<sup>1</sup> This kind of development was made possible by soliciting assistance for the more complex parts of development and to provide some quality assurance from a colleague of ours named Andrei Ghigea. As shown by my partner's research, Tudor Bunescu, such a game may be able to have a profound impact by helping to train dyslexic children in order to compensate for their difficulties especially when it comes to spelling and reading. Whereas a traditional approach leads to kids being bored, games provide a far more intuitive method for learning since kids may see it as less of a burden and instead interpret it as fun - motivating them to keep playing which intensifies the learning process - leading to better engagement, participation, achievement and recall of their learning.<sup>2</sup> We believe these reasons help keep our hypothesis reasonable and well-justified, hopefully paving the way towards an interactive way to help improve literacy even in very young children as allowed by our streamlined interface.

## **The learning aspect**

Our game revolves around using specially illustrated obstacles indicating checkpoints which deliver our learning payload; a mechanism that retrieves a word from a list made up of 496 common words which is read from an array and is stored into a string variable. This serves not only to improve the children's condition but also to potentially improve their vocabulary, allowing for a more rounded education and potentially even appealing in the future to a wide market segment.<sup>3</sup> Afterwards, the game splices the word into individual letters and reorders them randomly, soliciting the user to drag and drop the letters in such a way that they form the correct spelling of the word, which prompts the algorithm to check it against the original word and decide, upon successfully validating the user's spelling of the word against the one found in the list, whether to allow the user to continue or not. This also provokes the user with a challenge; words may have the same letters which greatly enhance the difficulty of the game and prevents it

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<sup>1</sup> "Top 15 Reasons to Use Unity 3D for Mobile Game Development." 2022. Wwww.expertappdevs.com. June 22, 2022. Accessed January 29, 2024. <https://www.expertappdevs.com/blog/15-reasons-to-use-unity-3d-for-mobile-game-development>.

<sup>2</sup> "Kids Are Here to Play - the Importance of Games | Cambridge English." n.d. Wwww.cambridgeenglish.org. Accessed January 29, 2024.

<https://www.cambridgeenglish.org/learning-english/parents-and-children/your-childs-interests/kids-are-here-to-play-the-importance-of-games/#:~:text=Our%20research%20shows%20that%20with>.

<sup>3</sup> "List of 5 Letter Words | Excel PDF CSV - CopyLists.com." 2022. February 25, 2022. Accessed January 29, 2024. <https://copylists.com/words/list-of-5-letter-words/>.

from inducing users into a state of boredom leading to disengagement and an abrupt decline in concentration leading to a reduced efficacy of the program's benefits for dyslexia.

### **Making the game fun**

We have decided to base the entertaining aspect of our game on “Flappy Bird” due to its proven effectiveness in audience retention and engagement which ensures that it can appropriately allow children to interact with it, whilst also being on the easier side to code for since it is 2D and many open-source assets and tutorials can be found for it, which while we may not have followed everything in them, they served as an excellent benchmark and source of information in the event that we needed to troubleshoot our code or implement a new function.<sup>4 5</sup> The player can use the spacebar or the touch screen in order to easily interact with the bird and keep it in the air as it flies through gaps in between pillars and floating objects. In the top left corner, an indicator displays the score representing the total amount of pillars the player has successfully passed. If a player collides with an object, then the game is over and the final score is issued within the “game over” menu, allowing for a sense of competition and urging the player to keep playing in order to improve their performance in it. Moreover, whenever the player has exceeded their previous score, it gets saved and is displayed on the “Game Over” scene in order to motivate the player and remind them of their progress.<sup>6</sup> This keeps the player engaged which enables the learning aspect to be adequately implemented helping to insure it plays its part in helping to treat dyslexia.

### **Intuitive design**

The game displays the letters required for the spelling challenges which are represented by easy to understand colourful icons that have been edited in order to be repurposed to hide scrabble notations so that they are perfectly aligned with our game's requirements.<sup>7</sup> A bright blue bird was chosen to serve as the main sprite due to how easily it stands out from the rest of the scene.<sup>8</sup> Vibrant colours appeal to children and help to draw them in. This helps them to perceive them easier since young children's eyes are not yet fully developed, helping kids understand letters

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<sup>4</sup>Warren, Tom. 2014. “‘Flappy Bird’ Creator Breaks Silence, Says He Pulled Hit Game Because It Was ‘Addictive.’” The Verge. February 11, 2014. Accessed January 29, 2024.

<https://www.theverge.com/2014/2/11/5400488/flappy-bird-pulled-because-it-became-addictive>.

<sup>5</sup> Adam. 2021. “How to Make Flappy Bird in Unity (Complete Tutorial) 🐦🔵.” Www.youtube.com. November 10, 2021. Accessed January 29, 2024. <https://www.youtube.com/watch?v=ihvBiJ1oC9U>.

<sup>6</sup> BMo. 2022. “The Easiest Way to Make a High Score in Unity.” Www.youtube.com. April 15, 2022. Accessed January 29, 2024. <https://www.youtube.com/watch?v=Y7GjVFFSMuI>.

<sup>7</sup> Kenney. 2015. “Letter Tiles.” OpenGameArt.org. January 21, 2015. Accessed January 29, 2024. <https://opengameart.org/content/letter-tiles>.

<sup>8</sup> bevouliin.com. 2015. “Free Game Asset - Grumpy Flappy Bird Sprite Sheets.” OpenGameArt.org. July 9, 2015. <https://opengameart.org/content/free-game-asset-grumpy-flappy-bird-sprite-sheets>.

more clearly and drag and drop them more easily into place.<sup>9</sup> In particular - our choice of warm background colours contributes towards people with dyslexia finding it easier to read - our icons have yellow frames and our background ranges between yellow and orange.<sup>10</sup> Moreover, these colours are particularly beneficial as they reduce the interference of computer-emitted blue light on circadian rhythms, helping to make it a healthier experience.<sup>11</sup> Another important aspect is the ability to drag and drop them helps make interactions more intuitive and lessens the learning curve for easy usage of the program, shifting kids' attention to keep them focused on the important aspects of the game and reducing the need for extensive training, supervision and interventions for technical reasons.<sup>12</sup> Additionally, the game makes use of a variety of other techniques to bolster the emotional impact of the game including the usage of an OST from a royalty free website.<sup>13 14</sup> Furthermore the OST we have chosen is completely instrumental and is played at low volumes, which helps to increase the player's ability to focus and can help to calm them down too, allowing for a more pleasant experience.<sup>15 16</sup> In an attempt to increase the users' immersion, we have also added a parallax effect throughout the game with the background scrolling at a slower speed compared to the obstacles.<sup>17</sup> In addition to that, the game opens up into a simple menu which offers the ability to exit, instructions for playing in order for the user to be able to have a fluid experience and to be able to start the game in their own time by using the buttons available regardless of when the app opens.<sup>18</sup> Moreover, the usage of C# as a compiled language rather than an interpreted language allows for a much more efficient usage of system resources allowing for smooth performance even on relatively weak systems - especially with such a simple game which further draws away the user's attention and concerns from the app itself towards the actual activities - especially those designed to bolster the skills dyslexia

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<sup>9</sup> Pancare, Rachel. 2018. "How Do Bright Colors Appeal to Kids?" Sciencing. April 23, 2018. Accessed January 29, 2024. <https://sciencing.com/do-bright-colors-appeal-kids-5476948.html>.

<sup>10</sup> Rello, Luz, and Jeffrey P. Bigham. 2017. "Good Background Colors for Readers." Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility - ASSETS '17. <https://doi.org/10.1145/3132525.3132546>. Accessed January 29, 2024.

<sup>11</sup> "BLUE LIGHT IMPACT in CHILDREN." n.d. Accessed February 7, 2024. <https://infantsee.aoa.org/Affiliates/InfantSEE/Documents/Blue-Light-Impact-in-Children.pdf>.

<sup>12</sup> Crawshaw, Ceara. 2022. "Drag & Drop UX Design Best Practices." Pencil & Paper. October 25, 2022. Accessed January 29, 2024. <https://pencilandpaper.io/articles/ux-pattern-drag-and-drop/>.

<sup>13</sup> Feldbacher, Chris. 2019. "19 Best Websites to Download Royalty-Free Music for Games." Blog.felgo.com. October 1, 2019. Accessed January 29, 2024. <https://blog.felgo.com/game-resources/free-music-for-games>.

<sup>14</sup> Innes, Ruby. 2022. "We Asked 8 Video Game Music Composers What Makes a Great OST." Kotaku Australia. September 28, 2022. Accessed January 29, 2024. <https://www.kotaku.com.au/2022/09/we-asked-8-video-game-music-composers-what-makes-a-great-ost/>.

<sup>15</sup> Schempp, Theresa. 2023. "How Music Can Help You Study | UMGC." University of Maryland Global Campus. August 28, 2023. Accessed January 29, 2024. <https://www.umgc.edu/blog/how-music-can-help-you-study#:~:text=When%20it%20comes%20to%20studying>.

<sup>16</sup> McCulloch, Alex. 2018. "Classical Pop (Instrumental)." OpenGameArt.org. August 20, 2018. Accessed January 29, 2024. <https://opengameart.org/content/classical-pop-instrumental>.

<sup>17</sup> Mike. 2020. "How to Make Looping Backgrounds [Unity Tutorial]." Www.youtube.com. July 24, 2020. <https://www.youtube.com/watch?v=A5YSbqqr3sc>.

<sup>18</sup> Murat. 2023. "Make Your MAIN MENU Quickly! | Unity UI Tutorial for Beginners." Www.youtube.com. May 8, 2023. <https://www.youtube.com/watch?v=DX7HyN7oJJE>.

affects.<sup>19</sup> These aspects prioritise the effectiveness of the game and helps to make a steeper impact towards lessening dyslexia's grip under far less time.

## **Conclusion**

The combination of the elements above allow for a seamless and easy usage of the app which will hopefully be reflected in the users' improved reading and spelling performance. Moreover, the optimised design should result in a very positive user experience, compelling people to spread the word about the effectiveness of the app. This will also help collect a significantly larger volume of user feedback which may then, due to the fact that the app is written in a high-level language allowing for easier programming, be used to further improve the program and help it ensure that it can fulfil the research question positively and excel at its purpose<sup>20</sup>.

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<sup>19</sup>“Unraveling Python: The Power of an Interpreted Language.” n.d. Www.upgrad.com. Accessed January 29, 2024. <https://www.upgrad.com/tutorials/software-engineering/python-tutorial/why-python-is-interpreted-language/#:~:text=Python%20is%20an%20interpreted%20language>

<sup>20</sup>Altexsoft. 2021. “The Good and the Bad of C# Programming.” AltexSoft. October 29, 2021. Accessed January 29, 2024. <https://www.altexsoft.com/blog/c-sharp-pros-and-cons/>.

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